

The Last Stand: Documentation

Welcome to **The Last Stand**, a fast-paced, top-down 2D survival shooter game. Your goal is simple: fight through waves of increasingly difficult enemies, scavenge for helpful power-ups, and survive long enough to defeat the final boss.

This guide is designed for players and non-technical readers to understand how the game works, how to play it, and how to get it running on your computer.

Gameplay Overview

In *The Last Stand*, you are dropped into an arena where enemies spawn from the edges of the screen. You must dodge obstacles, eliminate threats, and manage your health to progress.

- **Life System:** You start the game with **4 lives**, which are displayed as heart icons in the top-left corner of your screen. If an enemy hits you, you lose a heart.
- **Progressing:** Each game consists of **5 waves** of enemies. Once you clear all enemies in a wave, the next wave begins.
- **The Final Boss:** The ultimate challenge awaits you at **Wave 5**. A massive Boss Enemy with its own dedicated health bar will appear and summon minions. Defeating this boss wins the game!

Controls

The game features **independent movement and aiming**. This means you can run away in one direction while shooting backward or sideways at the enemies chasing you.

Keyboard Layout

Action	Control Keys	How it Works
Move Around	↑ ↓ ← → (Arrow Keys)	Moves your character in 8 directions (including diagonals like up-and-right).
Shoot Straight	W, A, S, D Keys	Shoots bullets in the direction of the key (W for up, S for down, etc.).

Action	Control Keys	How it Works
Shoot Diagonally	Hold two shoot keys	Hold two adjacent keys together (e.g., W + D fires Northeast).
Pause Game	Pause Button	Click the button in the top-right corner to freeze the action.

Enemies

As you advance through the levels, the game gets tougher. The number of enemies increases every wave, and they spawn faster as you reach higher levels. You will encounter four types of threats:

- **Basic Enemy:** The standard monster. It moves directly toward you at a normal pace and takes 1 hit to defeat.
- **Speedy Enemy:** Much faster than normal enemies. You can spot them by their quick running and attacking animations.
- **Tanky Enemy:** A slower, heavily armored variant. These require **2 hits** to destroy.
- **The Boss Enemy:** Spawns exclusively on **Wave 5**. It has a large health bar at the top of the screen and will continuously summon minion enemies to protect itself.

Difficulty Note: The chance of encountering Speedy and Tanky enemies increases by 5% with every level you clear!

Power-Ups

Defeated enemies or levels will occasionally grant you power-ups. Walking over these items gives you an immediate advantage:

- **Fire Rate (Red):** Supercharges your weapon. Your weapon cooldown drops from 500ms down to 200ms, letting you shoot more than twice as fast. *(Lasts 5 seconds)*
- **Movement Speed (Blue):** Makes your character run significantly faster, allowing you to easily outrun mobs or escape tight corners. *(Lasts 10 seconds)*
- **Heal (Green):** Instantly restores 1 heart container. This will not grant extra health if you are already at your maximum of 4 lives. *(Instant)*

Game Screens & Saving

Menu Guide

The game features simple, dark semi-transparent menus that overlay the action when necessary:

- **Main Menu:** Start a brand new game, exit, or click **Continue** to load your last saved game.
- **Pause Menu:** Pauses the action. From here, you can resume playing, or choose to return to the Main Menu / Exit.
- **Game Over Screen:** If you lose all 4 hearts, you can choose to **Respawn** (which restarts you at the beginning of your current level) or return to the Main Menu.
- **Win Screen:** Celebrates your victory after defeating the Level 4 Boss!

How Saving Works

You don't have to worry about losing your place. The game **automatically saves your progress** whenever you pause and choose *Back to Main Menu*.

The game remembers:

- Your current Level and Wave
- How many remaining lives (hearts) you have
- Your exact position on the map

Important: Choosing New Game from the main menu, or returning to the main menu after a formal Game Over or Victory, will clear your save file so you can start fresh.

Sound and Music

The game features dynamic audio to keep you immersed:

- The game features distinct background tracks for the **Main Menu** and the active **Gameplay**.
- Sound effects (SFX) trigger dynamically when you shoot, get hit, or pick up power-ups.
- The system smoothly manages volume and prevents tracks from abruptly restarting or overlapping.

How to Install and Run the Game

Prerequisites

To play this game, your computer needs **Java 11 or higher** installed. The game does not require any heavy graphics cards or third-party platforms like Steam—it runs entirely on standard computer frameworks.

Launching the Game

Depending on how the game was delivered to you, use one of the options below to play:

Option A: The Easy Method (If provided an executable)

Double-click TheLastStand.exe (Windows) or TheLastStand.jar (Mac/Linux) in the main folder. The game will automatically open in a distraction-free, full-screen mode scaled to your monitor's size.

Option B: The Command Line Method (For testing)

If you are launching the game via a terminal or command prompt, follow these steps:

1. Open your terminal application and navigate to the project folder.
2. Compile the game files by typing:

Terminal:

```
javac -d TheLastStand/out  
TheLastStand/src/TheLastStand.java  
TheLastStand/src/fileio/*.java  
TheLastStand/src/gameloop/*.java  
TheLastStand/src/objects/*.java  
TheLastStand/src/sound/*.java  
TheLastStand/src/ui/*.java
```

Run the game using:

Terminal:

```
java -cp "TheLastStand/out;TheLastStand" TheLastStand
```

Note: Always make sure the game folder contains the asset/ folder (which holds the sounds and images) in the same directory, otherwise the game will load without graphics or music!